Teacher-Calibrated Perceptual Diversity for One-Step Diffusion

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Abstract

We investigate direct one-step text-to-image generation with a fixed rank-32 Stable Diffusion student and a 36,000-step training budget. Our DMD-based proposal combines distribution matching, perceptual regression and matching the teacher's variation across noise seeds for the same caption. Evaluation uses all 5,000 COCO-2017 validation images and matched captions and noise. The completed proposal achieves FID 30.40 and CLIP 31.12 at one denoiser evaluation, compared with FID 37.02 and CLIP 30.57 for our DMD baseline. The perceptual-only ablation has lower FID; adding diversity matching trades FID for a small increase in pairwise perceptual distance. We provide measured baseline results, selected qualitative examples and explicit experimental limitations.

1. Introduction

Diffusion models generate detailed images but require repeated denoiser evaluations. Distillation transfers a pretrained teacher into a generator with a smaller inference budget. Our goal is a practical one-step student that preserves visual detail and useful variation under a limited adapter budget. A low reconstruction loss alone is insufficient: blurry or repetitive images can fit a restrictive target without capturing the teacher distribution.

We study whether perceptual regression and teacher-calibrated within-prompt variation help a DMD-based LoRA student. The proposal matches the perceptual distance between two student images to the distance between their paired teacher outputs. This target avoids rewarding image separation without a teacher reference. The contribution is a controlled image-level extension under a fixed backbone, student capacity and total training-step budget.

2. Related work

Latent diffusion enables high-resolution generation using a pretrained VAE and a denoiser operating on compressed image latents [6]. LoRA reduces the number of trained parameters [3]. Progressive distillation repeatedly replaces two teacher transitions with one student transition [7]. Consistency models learn predictions that agree along a probability-flow trajectory [8]. InstaFlow uses reflow to straighten noise–image couplings before one-step distillation [4]. These objectives provide complementary baselines for a common one-step inference budget.

DMD combines distribution matching with regression to teacher-generated outputs [9]; SwiftBrush trains a generator using variational score distillation [5]. Perceptual regression is already part of original DMD. Relational diffusion distillation also studies cross-sample information [2]. We therefore evaluate a specific same-caption teacher-distance objective in our restricted setting, without claiming the first use of perceptual or relational distillation. Our baseline implementations use reduced-capacity adapters and differ from the published full-model recipes; their measurements are local experimental results.

3. Method

One-step generator and distribution matching. The student $G_\theta(z, c)$ maps Gaussian noise z and a caption c to a clean VAE latent in one conditional U-Net call. Teacher and fake-score networks supply clean estimates $\hat{x}_0^T$ and $\hat{x}_0^F$ from a noisy student latent. The DMD gradient direction is

$$g = \frac{\hat{x}_0^F - \hat{x}_0^T}{\mathrm{mean}(|G_\theta - \hat{x}_0^T|) + \epsilon}. \tag{1}$$

The implementation injects the stopped direction into the generator and fits the auxiliary scorer with epsilon denoising on detached student outputs. Both generator and scorer use rank-32 adapters. The VAE and text encoder remain frozen. Guidance is incorporated during training; student inference uses a single conditional branch with no additional CFG pass.

Teacher-calibrated perceptual objective. Let $I_i = D(G_\theta(z_i, c))$ and $J_i = D(T(z_i, c))$, where D is the VAE decoder and T denotes a cached teacher trajectory. RGB values are clamped to $[-1, 1]$ for frozen LPIPS-VGG dis-



Figure 1. Selected qualitative examples from completed 36,000-step students. Each row uses the identical recorded caption and initial evaluation noise across methods. All students use one step / one NFE; the teacher is a 64-step / 128-NFE reference. SwiftBrush uses teacher CFG 4.5 and FP32; other students use CFG 7.5 and BF16. The selected subjects are a dog in flowers, teddy bears and a lighthouse. Dog/lighthouse use fixed preset prompts; the teddy-bear row is a validation example. Full-set metrics use all 5,000 validation outputs. Baseline names denote our LoRA adaptations.

tance d [10]. We use

$$L_{\text{reg}} = \tfrac{1}{2}\sum_{i=1}^{2} d(I_i, J_i), \tag{2}$$

$$L_{\text{div}} = \left[d(I_1, I_2) - \text{sg}(d(J_1, J_2))\right]^2, \tag{3}$$

$$L = L_{\text{DMD}} + 0.25 L_{\text{reg}} + 0.1 L_{\text{div}}. \tag{4}$$

The two noise seeds are independent, while the caption is shared. Both student branches receive gradients. Teacher outputs and LPIPS weights are frozen; decoder backpropagation supplies gradients to the student latent. Only the student denoiser and decoder are needed at inference.

Controlled ablations. Four variants cross latent-MSE versus LPIPS regression with diversity weight $\lambda_d \in \{0, 0.1\}$. They share the exact same DMD initialization, teacher-pair bank, captions, optimizer state, training seed and effective batch size. Each follows a 36,000-step pipeline: the common DMD initialization phase accounts for half the budget, followed by the loss-specific phase. The grouped latent-MSE control is required to separate changes in the anchor protocol from the effects of the proposed losses.

4. Experimental protocol

Data and evaluation coverage. We use SD1.5 with 512×512 outputs. COCO-2017 captions are ordered by image ID, selecting the lowest-ID caption per image [1]. Training uses disjoint pools of 50,000 and 68,287 training-image captions. Together these cover the 118,287 available training images. The proposed DMD-family objectives use captions, random noise and generated teacher anchors, with no ground-truth image supervision in those losses. All 5,000 official validation images are held out from student training. Reference images use bicubic resize and center crop. Evaluation covers every official validation image with its selected caption.

Every final quality evaluation generates one image for each of the 5,000 validation captions, using the same evaluation noise stream (seed 12345). FID and Inception Score (IS) measure distribution-level generation quality. CLIP uses ViT-B/32 and reports mean $100\max(\cos(I, c), 0)$. These are full COCO-2017 validation results under our caption/preprocessing protocol, which differs from published COCO-2014 benchmark protocols.

Capacity, updates and compute. All students have rank-32 LoRA generators with 6.38 million trainable parameters, effective batch 16, training seed 42, and 36,000 generator updates. All student comparisons use one step / one NFE. Our DMD study uses microbatch two with eight accumulation passes, AdamW, generator/scorer learning rates 10^{-4}, 100 warmup steps, EMA 0.995 and one scorer update per generator update. Its shared bank has 5,000 teacher images from 2,500 prompts with two noise seeds each. The frozen teacher uses 64 DDIM steps and CFG 7.5, costing 640,000

Method (36k steps)	NFE	FID ↓	IS ↑	CLIP ↑
Progressive distillation	1	49.96	23.12	29.10
Consistency distillation	1	83.80	16.82	28.33
DMD	1	37.02	29.38	30.57
InstaFlow	1	51.40	21.85	29.06
SwiftBrush	1	40.72	25.46	29.38
Ours	1	**30.40**	**33.76**	**31.12**
Teacher reference: 64 steps, CFG 7.5				
SD1.5 teacher	128	24.10	34.46	31.30

Table 1. Full-validation quality: 5,000 generated images and 5,000 real references. Student rows use the full 36,000-step budget. Bold marks the best student value within this comparison. SwiftBrush uses teacher CFG 4.5/FP32 and other students use CFG 7.5/BF16; the teacher has a longer trajectory. All baseline names refer to our implementations.

denoiser evaluations to create the shared bank once. Decoder and LPIPS gradients add training compute; equal generator updates do not imply equal total compute.

PD uses the epsilon-backbone progressive schedule; consistency uses a VP boundary blend and latent L2. DMD uses normalized score matching with latent regression, while InstaFlow uses the VP-to-velocity adaptation and cached reflow couplings. SwiftBrush retains a rank-32 student with its recorded optimizer/scorer settings, teacher CFG 4.5 and FP32 precision. Other students use teacher CFG 7.5 and BF16. All methods appear in one comparison, with these recorded settings disclosed. Concurrent GPU workloads prevent controlled latency claims; NFE provides the inference-budget comparison.

Diversity and visual selection. Diversity is measured separately on 128 held-out captions with four fixed seeds each, averaging the six within-caption LPIPS-Alex distances. AlexNet differs from the VGG training metric. This 512-image diversity test is distinct from the full 5,000-image quality evaluation. Figure 1 uses three selected subjects with identical row captions and noise across methods. Two use preset prompts; the teddy-bear example comes from validation.

5. Results and analysis

The full proposal obtains FID 30.40, IS 33.76 and CLIP 31.12 at one NFE after 36,000 total training steps. Its FID is 6.62 lower (17.9 percent) than the DMD baseline at the same total step budget. This comparison changes both the regression objective and anchor grouping. Against the matched grouped control, the FID difference is -3.80. Perceptual regression alone obtains FID 29.94. Adding diversity matching changes FID by +0.46, while mean pairwise LPIPS-Alex distance changes from 0.7086 to 0.7129. The added term does not improve FID in this paired comparison.

Variant	Regression	λ_d	FID ↓	Diversity
Grouped DMD control	Latent MSE	0	34.20	0.7283
Perceptual only	LPIPS	0	**29.94**	0.7086
Diversity only	Latent MSE	0.1	33.98	0.7274
Ours	LPIPS	0.1	30.40	0.7129

Table 2. Factorial study with a shared initialization and 36,000 total training steps. Diversity is pairwise LPIPS-Alex distance across four seeds for each of 128 prompts. Bold marks the lowest ablation FID; a larger perceptual distance does not itself imply higher quality.

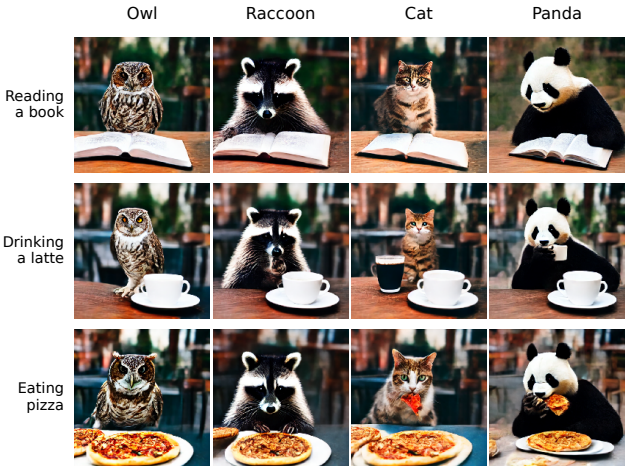


Figure 2. Selected compositional prompt-control grid from the 36,000-step proposal at one step / one NFE. Prompts follow 'A DSLR photograph of a {animal} {action}, sitting at an outdoor cafe table.' All cells share initial noise seed 12546. The displayed grid is manually selected for visual quality and animal/action adherence from a recorded candidate seed pool. This is discrete prompt composition.

With latent regression, diversity matching changes FID by -0.22 and pairwise LPIPS from 0.7283 to 0.7274. A larger perceptual distance alone does not establish higher visual quality or semantic diversity.

Visual findings and failure modes. The selected dog and lighthouse examples illustrate coherent subjects and scene structure, rather than typical-case performance. The teddy-bear row also exposes limited color variation despite plausible texture. Other saved samples expose limitations: the proposal omits the person in a dining-room caption, does not reliably separate requested teddy-bear colors, and renders a requested upside-down stop sign upright. Distribution-level FID and CLIP improvements therefore do not establish reliable counting, spatial relations or complete prompt adherence. The teacher reference also uses a much larger inference budget.

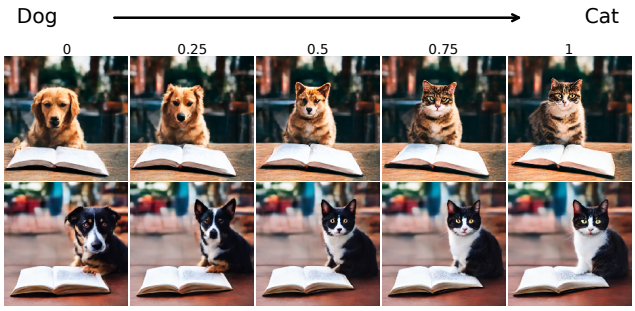


Figure 3. Text-embedding interpolation from dog to cat with the 36,000-step proposal at one step / one NFE. Endpoint prompts use 'A DSLR photograph of a {animal} reading an open book, sitting at an outdoor cafe table.' Columns linearly interpolate the CLIP hidden states at the labelled weights; each row keeps the same initial noise. Two rows are selected from three candidate seeds.

6. Limitations and conclusion

Resource and training-budget constraints. Available GPU resources and project budget limit each student to 36,000 generator updates, effective batch 16, rank-32 adapters and one training seed. This amounts to 576,000 samples in the generator training stream. Published training budgets vary: SwiftBrush reports approximately 65,000 updates at batch 64 [5], while the original consistency-model experiments use 600,000–1,000,000 iterations across different datasets and training variants [8]. These settings differ in model capacity, batch size and training task, so raw iteration counts do not equalize compute. Our constrained runs may leave both the proposal and baselines under-optimized; longer training could change their relative performance. The results establish a comparison within our budget, rather than the performance ceiling of the published methods.

The completed results support stronger full-validation quality for the combined proposal than our DMD baseline at a matched total update budget. The factorial study favors perceptual regression alone for FID; the added diversity term does not establish a consistent benefit. One training seed provides no estimate of run-to-run uncertainty. Adapter capacity, teacher-pretraining overlap, the restricted regression bank and validation reuse during development limit generality. Perceptual distance can increase through artifacts and cannot guarantee semantic diversity. Broader claims require additional seeds and controlled compute measurements.

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